

ES -116

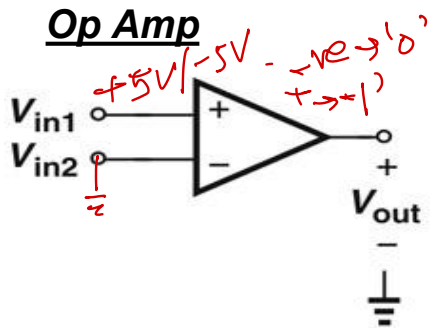
Principles and Applications of Electrical Engineering

Semester II, 2025-26

More functionalities with OpAMP

Recap

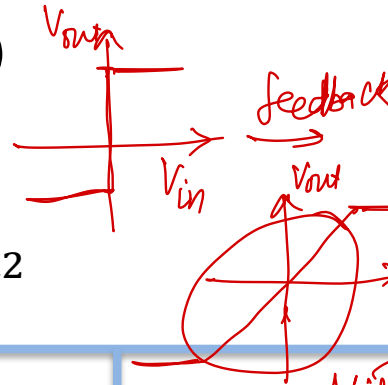
Op Amp



$$V_{out} = A_0(V_{in1} - V_{in2})$$

$$V_{in1} - V_{in2} = V_{out}/A_0$$

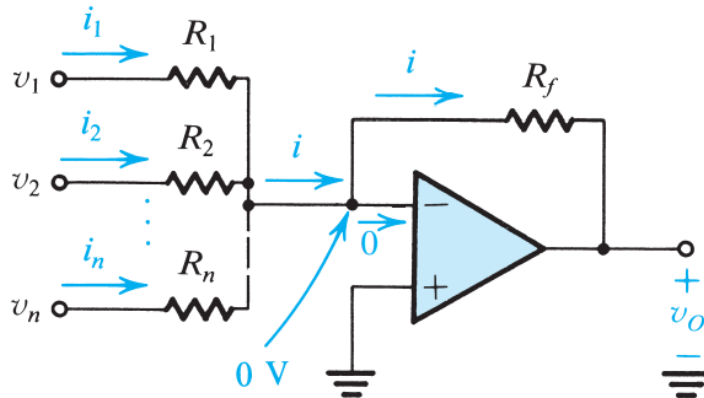
$$\text{If } A_0 \rightarrow \infty, V_{in1} \approx V_{in2}$$



Virtual short

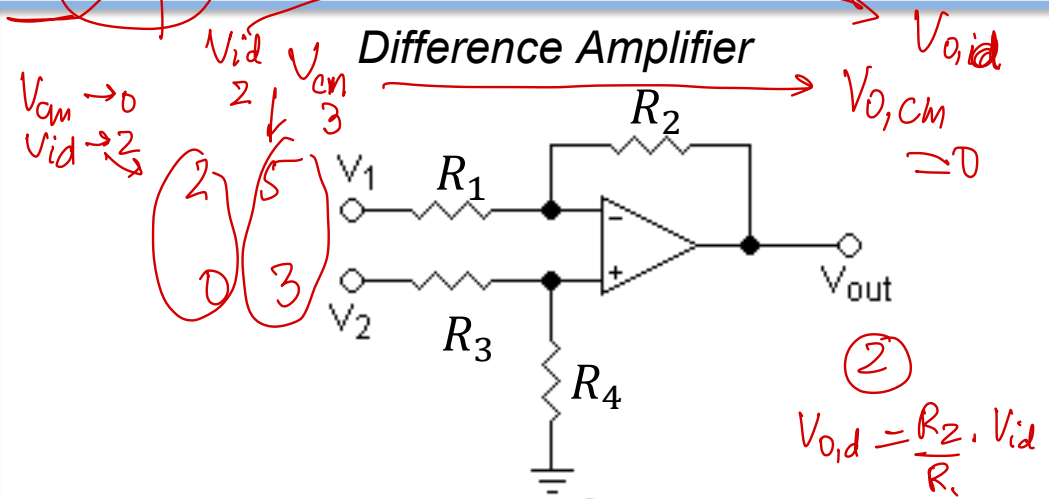
Due to infinite gain of op amp, V_{in2} appears to same as V_{in1} .

Voltage Summing Inverting Amplifier



$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \dots + \frac{R_f}{R_n}v_n\right)$$

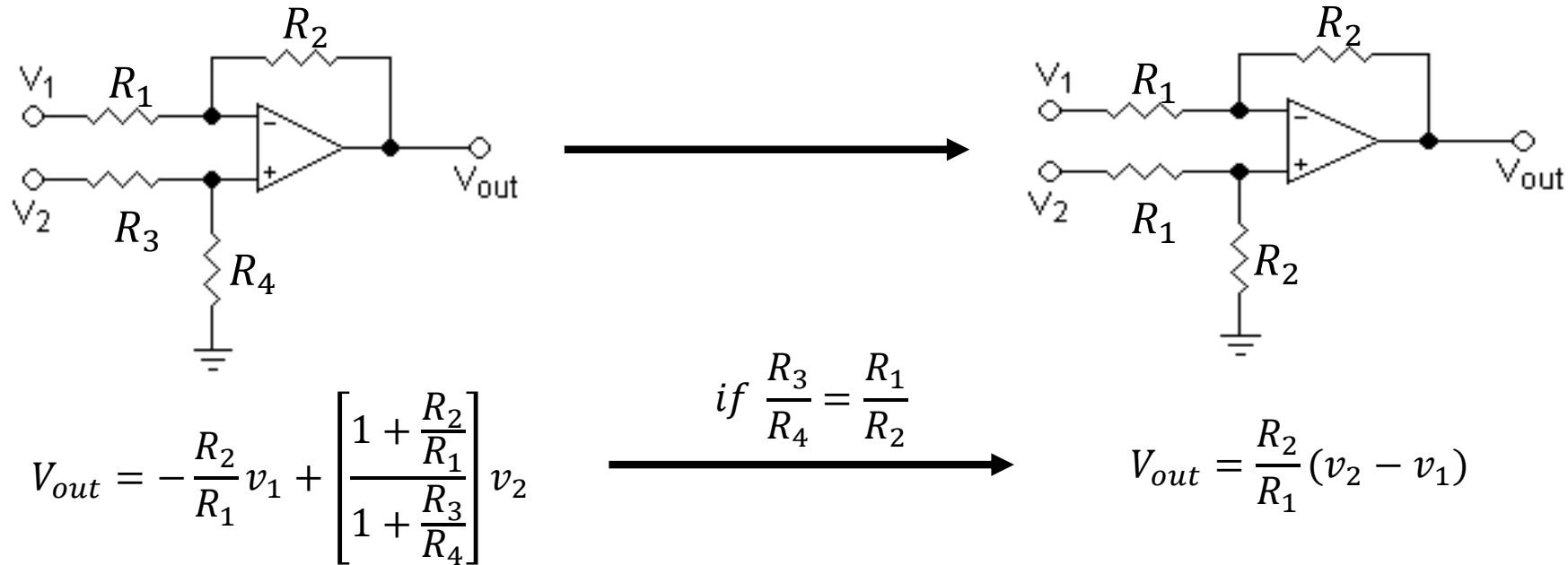
Difference Amplifier



$$V_{0,d} = \frac{R_2}{R_1} \cdot V_{id}$$

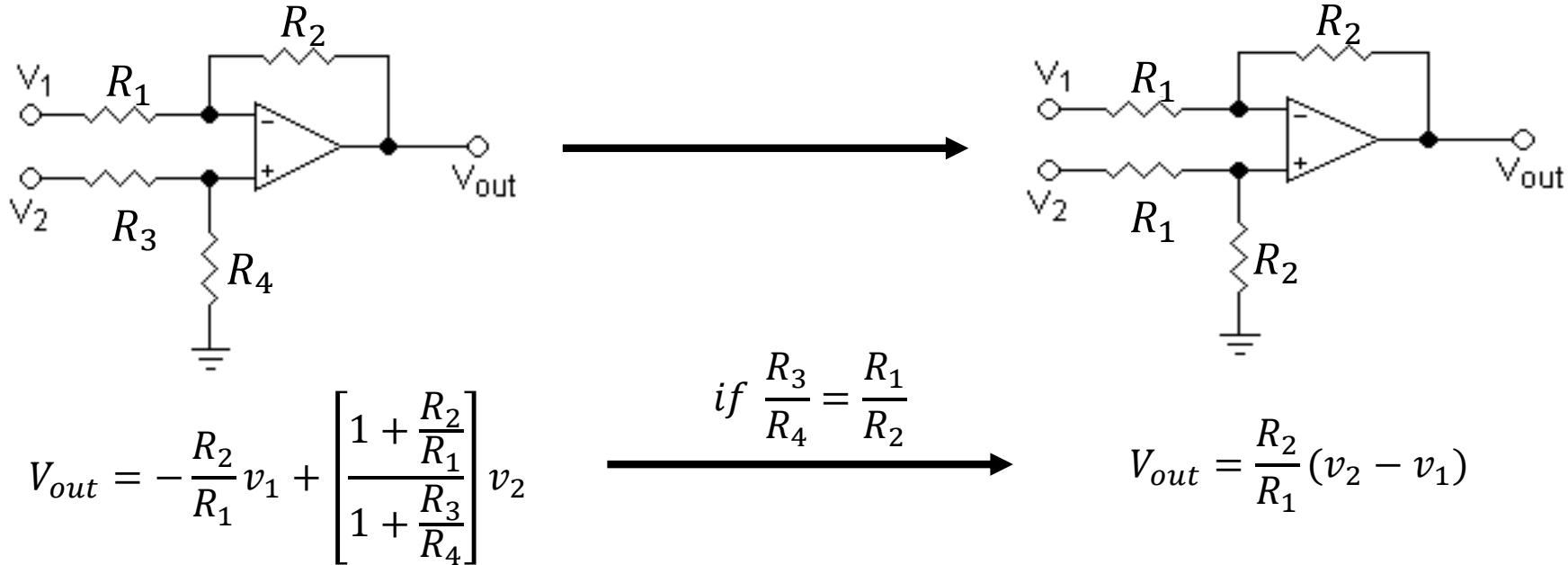
$$V_{out} = -\frac{R_2}{R_1}v_1 + \left[\frac{1 + \frac{R_2}{R_1}}{1 + \frac{R_3}{R_4}} \right] v_2 \xrightarrow{\text{if } \frac{R_3}{R_4} = \frac{R_1}{R_2}} V_{out} = \frac{R_2}{R_1}(v_2 - v_1)$$

Recap (Difference Amplifier)



What is the meaning of the differential and common-mode signals and corresponding gain?

Recap (Difference Amplifier)



Generic form: $V_{out} = A_d V_{id} + A_{cm} V_{cm}$

A_d : Differential gain, $v_{id} = v_2 - v_1$

A_{cm} : Common mode gain, $V_{cm} = V_{noise}$

$$A_d = \frac{V_{o,d}}{V_{id}}$$

$$A_{cm} = \frac{V_{o,cm}}{V_{i,cm}} = 0$$

$$V_{id} = 10V$$

$$V_{o,cm} = 100\mu V$$

$$\frac{A_d}{A_{cm}} = 10^5$$

$$\Rightarrow A_{cm} = 10^{-5} \times A_d$$

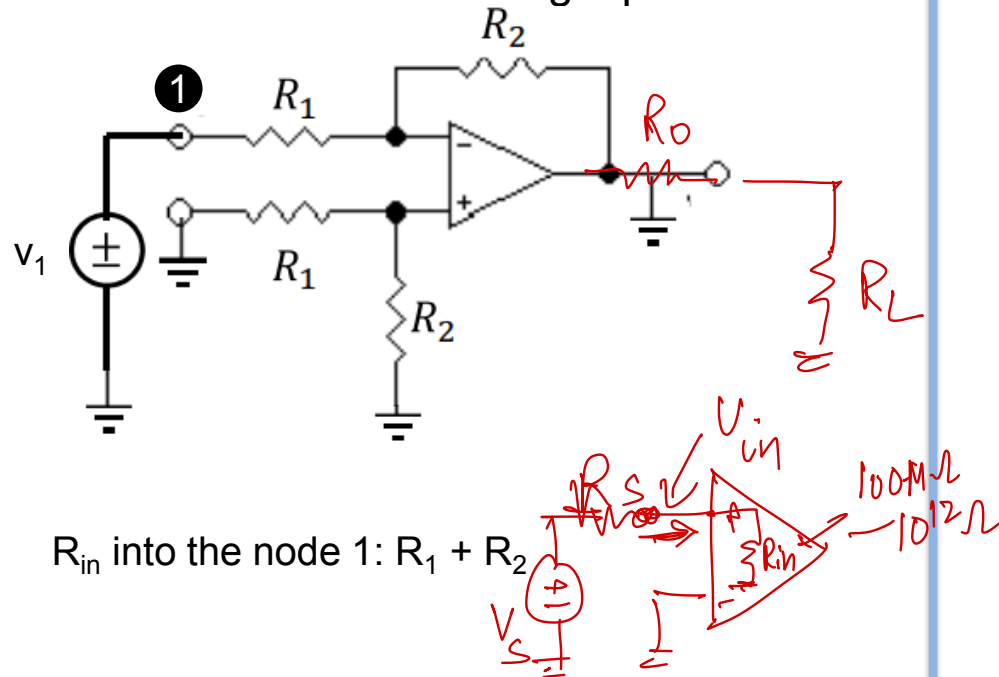
$$CMRR = 100dB \quad CMRR \rightarrow \infty$$

CMRR is a metric used to quantify the ability of the device to reject common-mode signals.

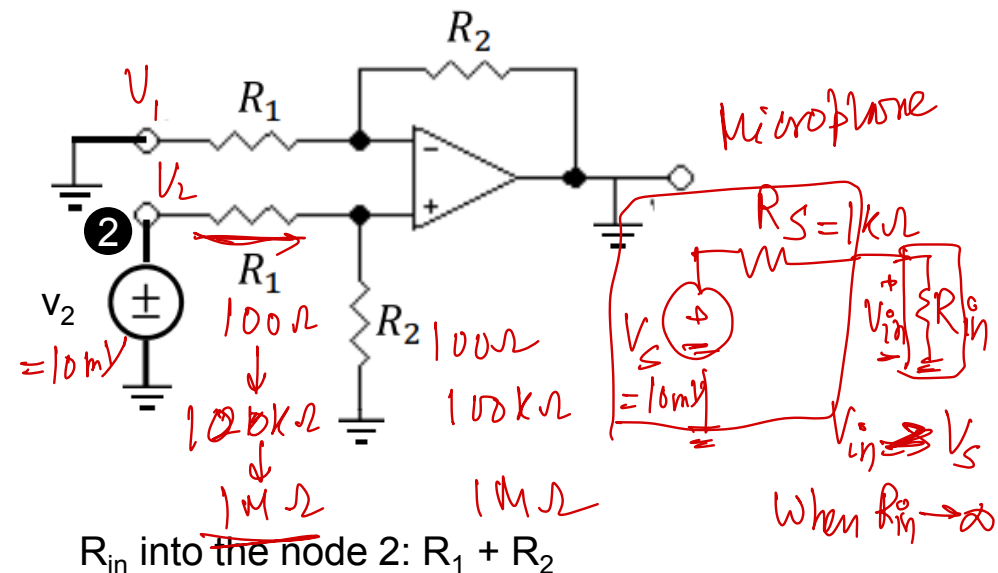
$$CMRR = 20 \log \frac{|A_d|}{|A_{cm}|}$$

Issue of input impedance

Input resistance into the inverting input terminal



Input resistance into the non-inverting input terminal

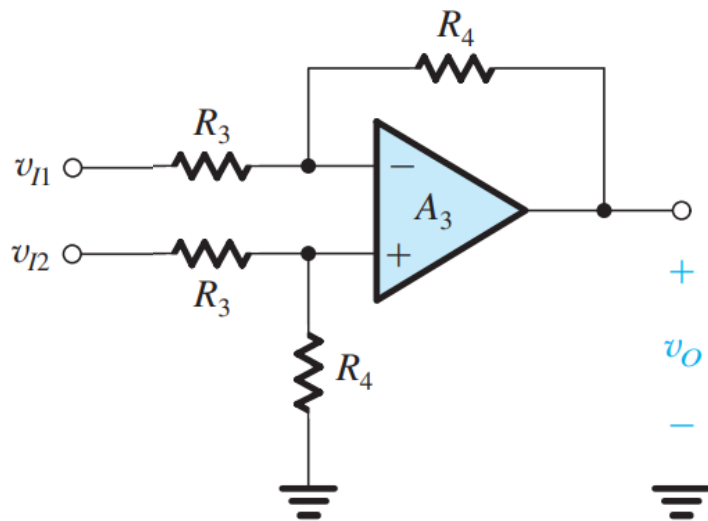


Finite input resistance can result in loading at the source end.

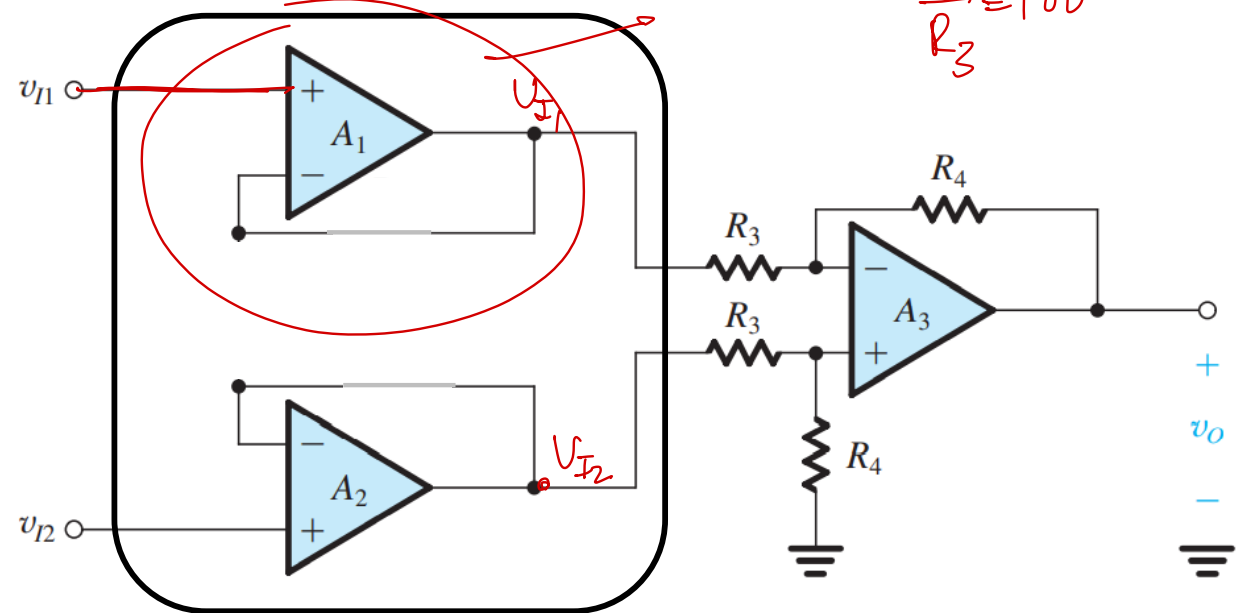
What solution do you propose other than choosing a large value of R_1 and R_2 ?

Solution to deal with loading at Input

Difference amplifier

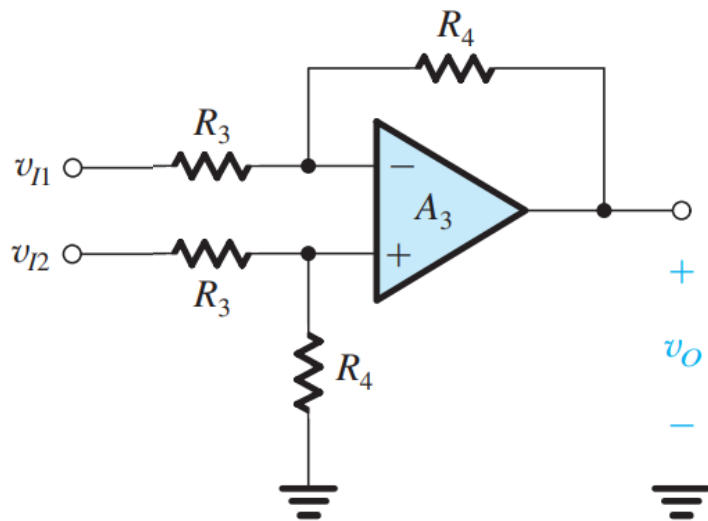


Using voltage buffers at each input?

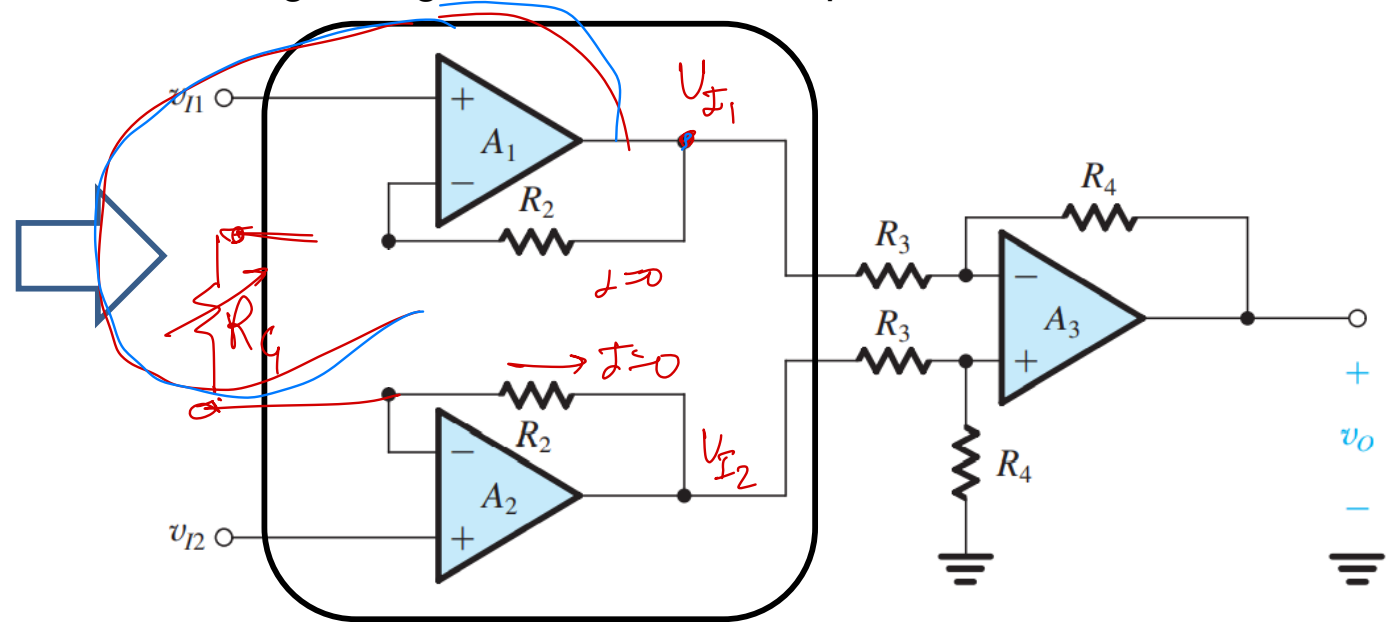


Is it possible to extract gain from this stage?

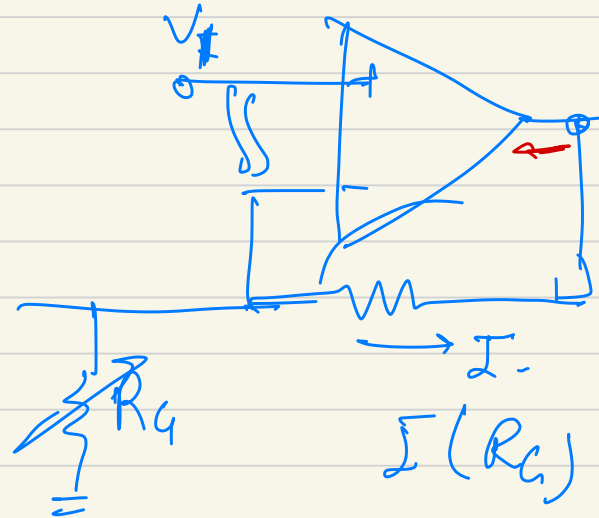
Difference amplifier



Using voltage buffers at each input

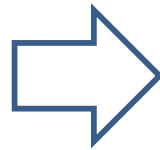
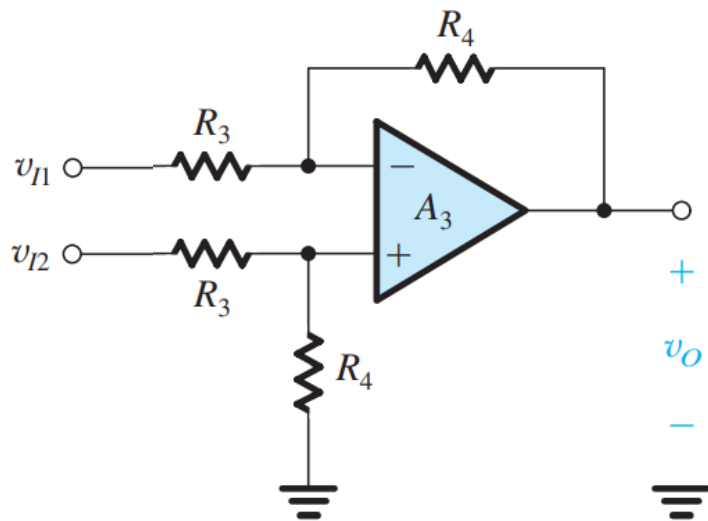


→ Gain setting resistor turn the simple voltage follower into an non-inverting amplifier because no current flows through feedback without R_G , while you can tune the current through feedback resistor using R_G , thus changing the output voltage.

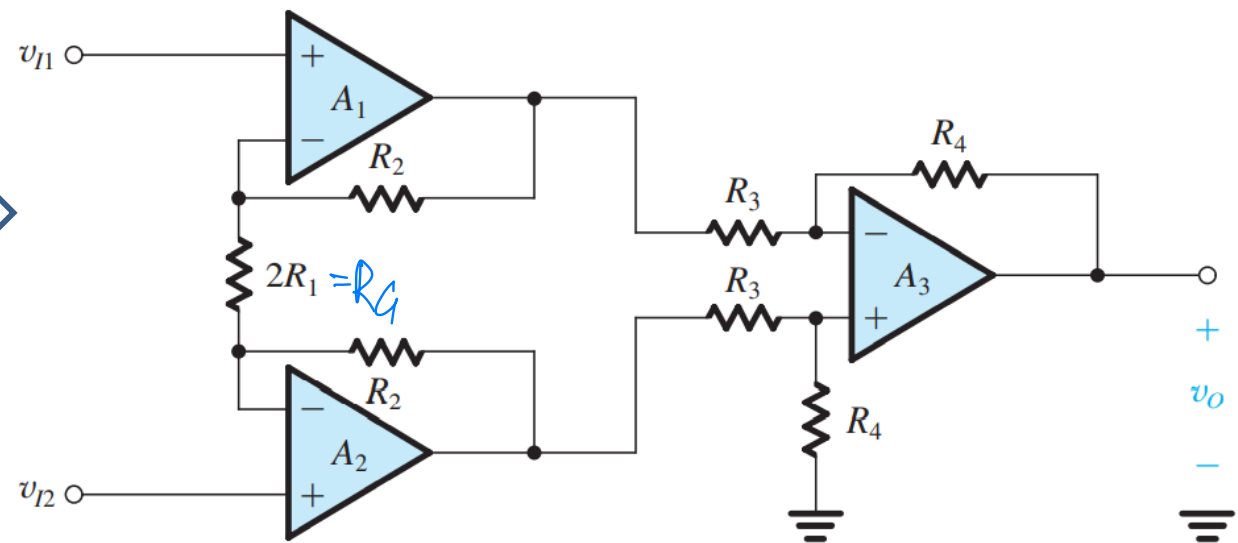


The grand solution

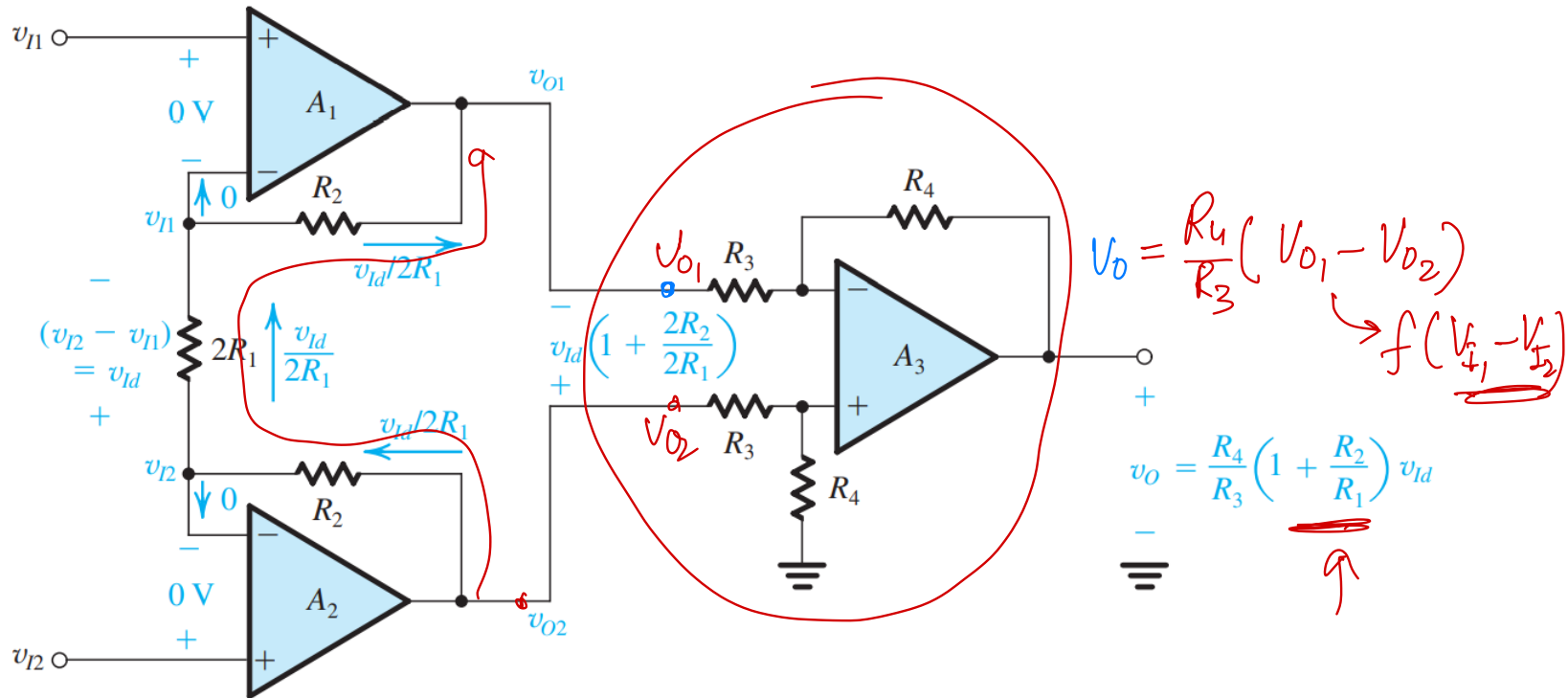
Difference amplifier



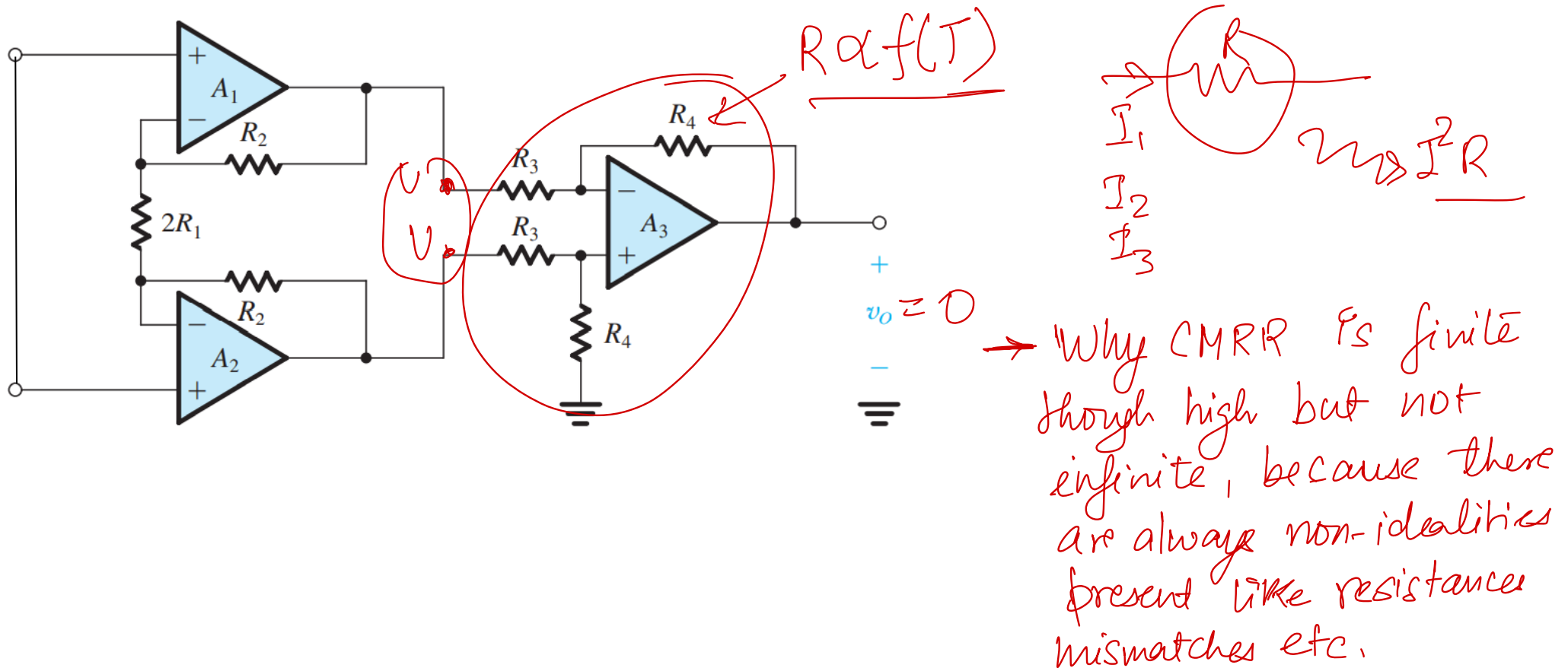
Instrumentation amplifier



Instrumentation amplifier: Differential gain

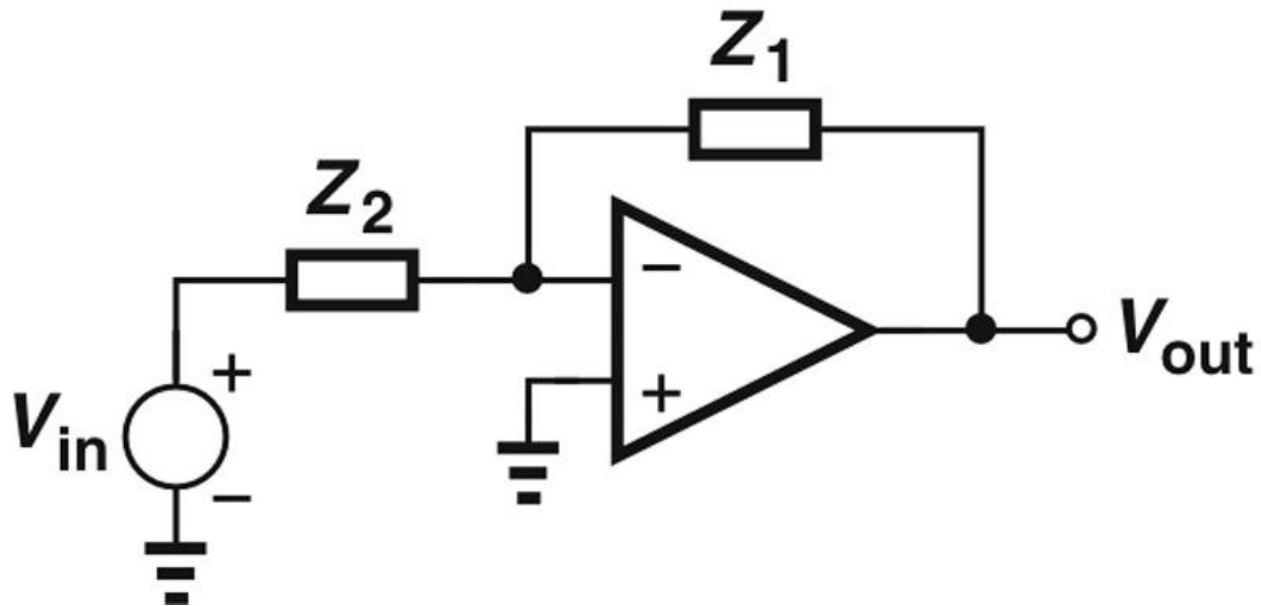


Instrumentation amplifier: Common-mode gain



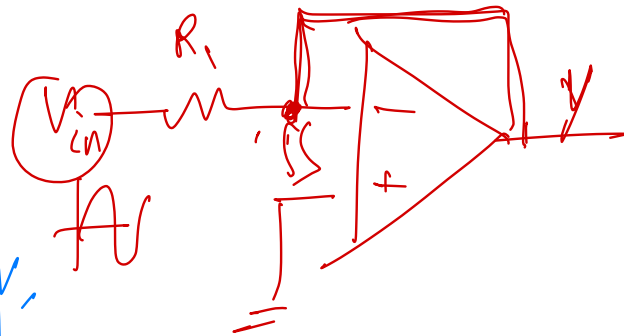
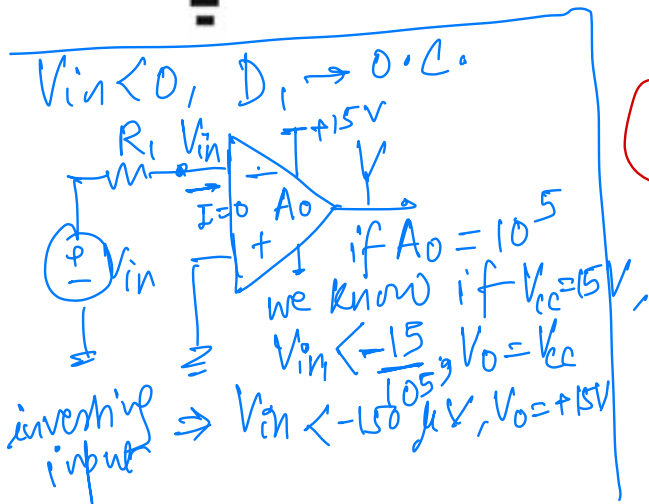
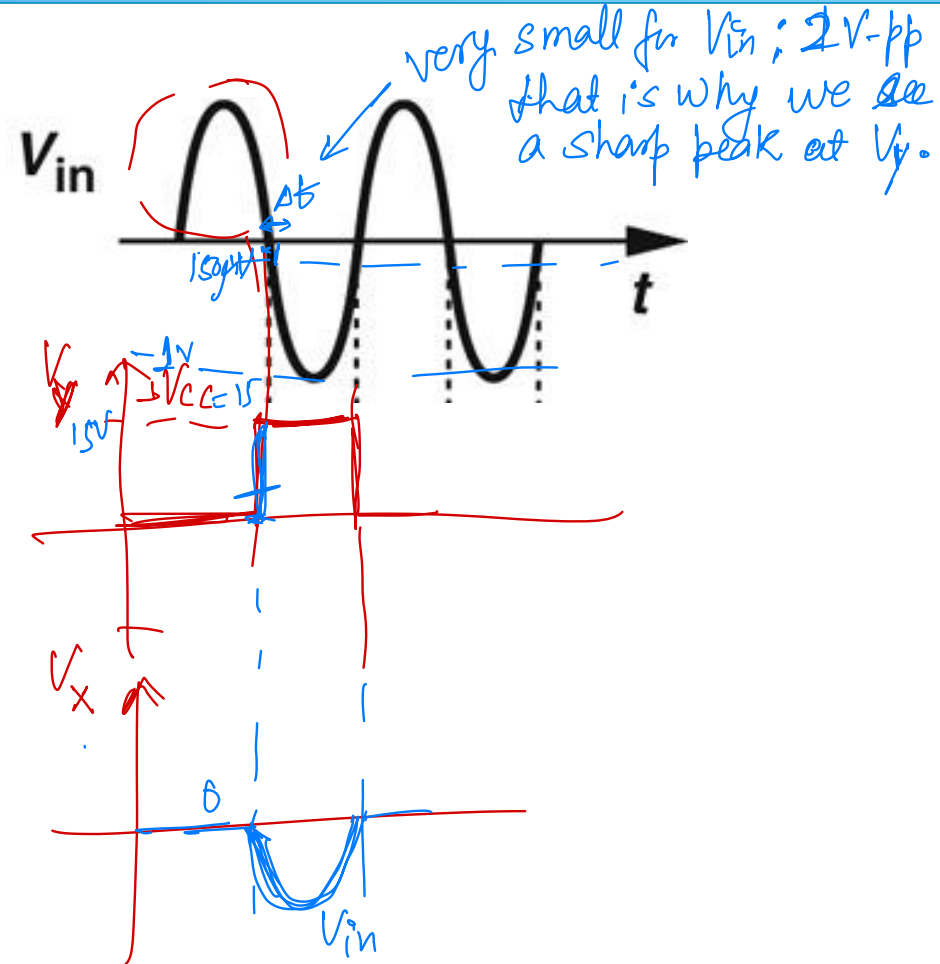
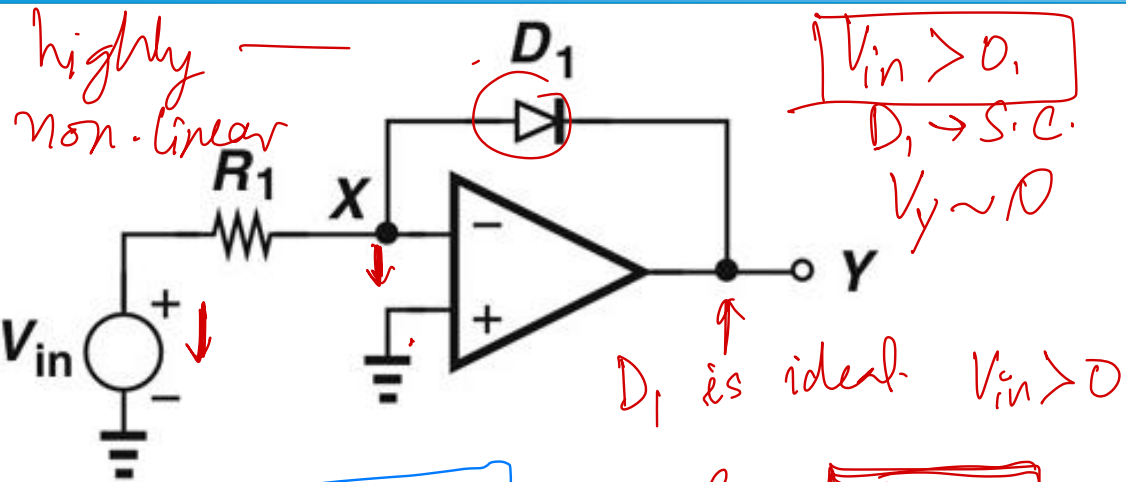
More functionalities with OpAmp

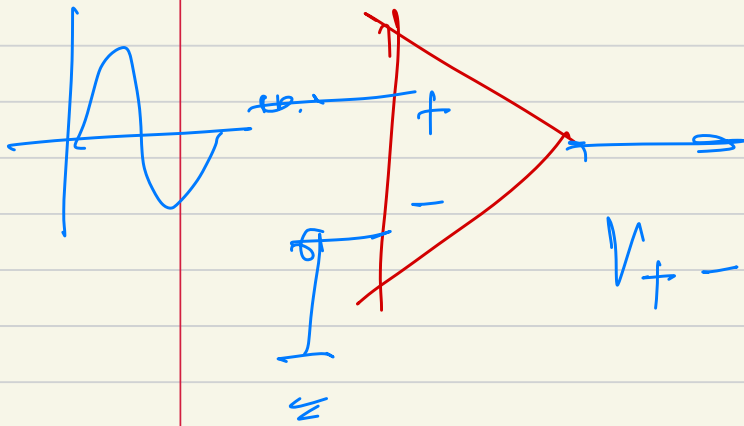
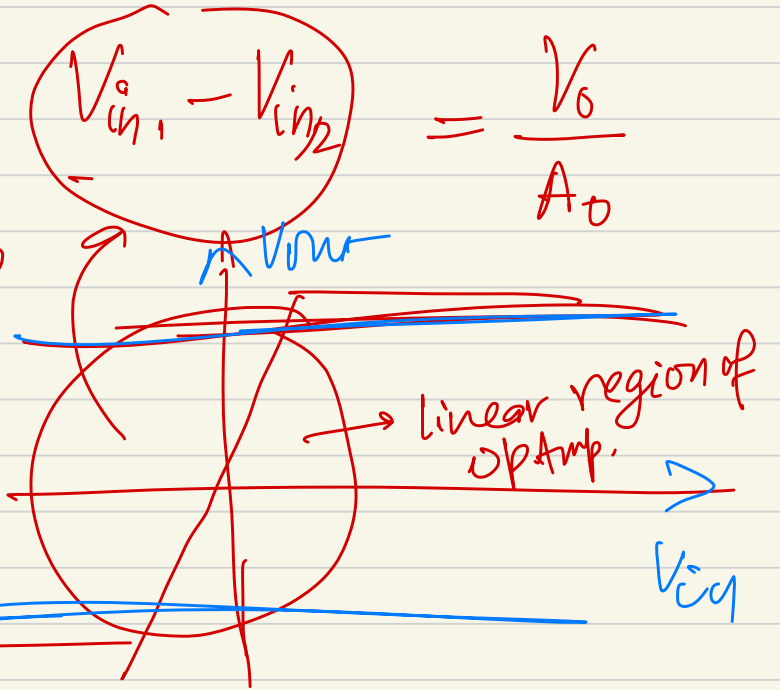
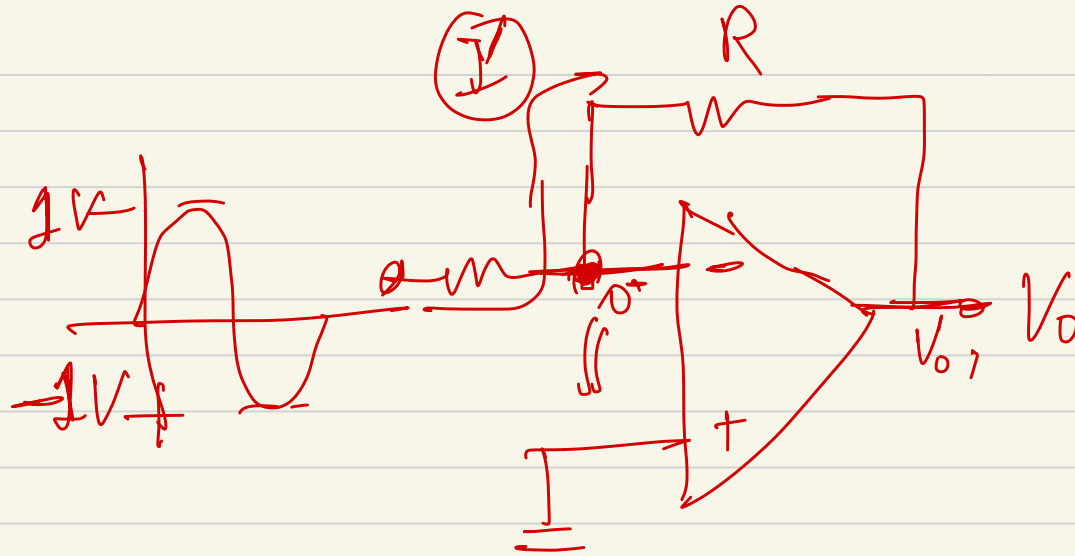
Inverting configuration with complex impedances



- Here, Z_1 , Z_2 could be any of the other electronic components such as resistors, capacitors, inductors, diodes, or transistors.

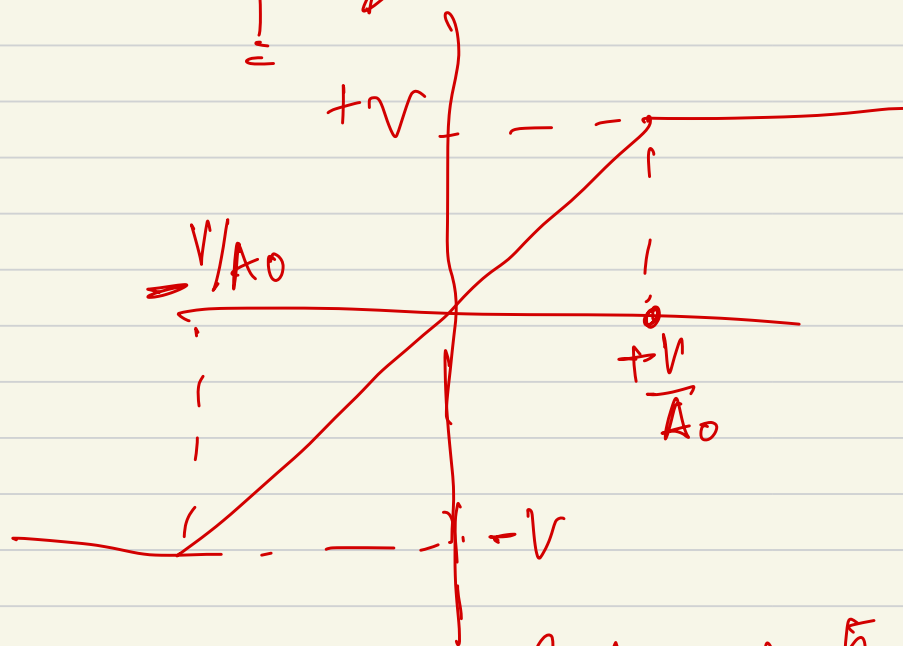
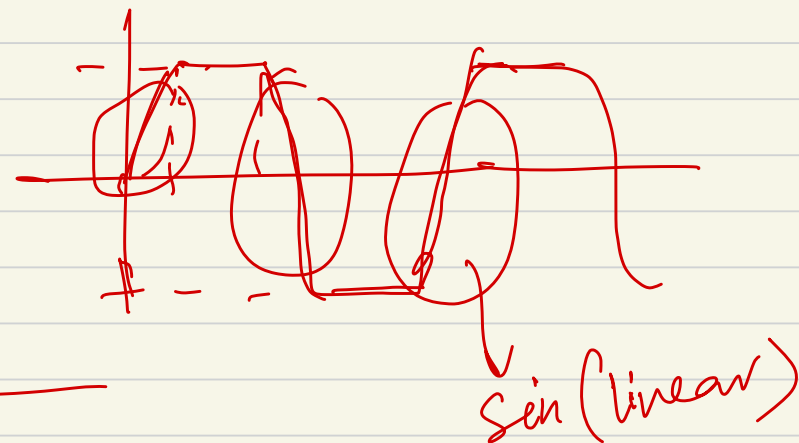
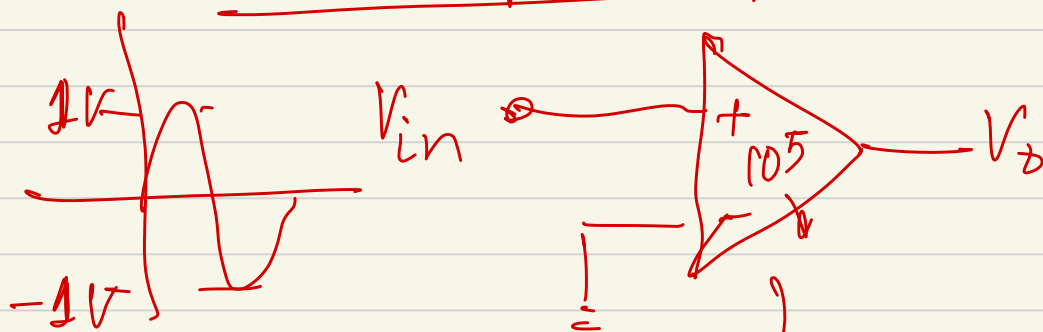
"Known" functionalities





$$V_+ - V_- \rightarrow V_o = \begin{cases} +15 \\ -15 \end{cases}$$

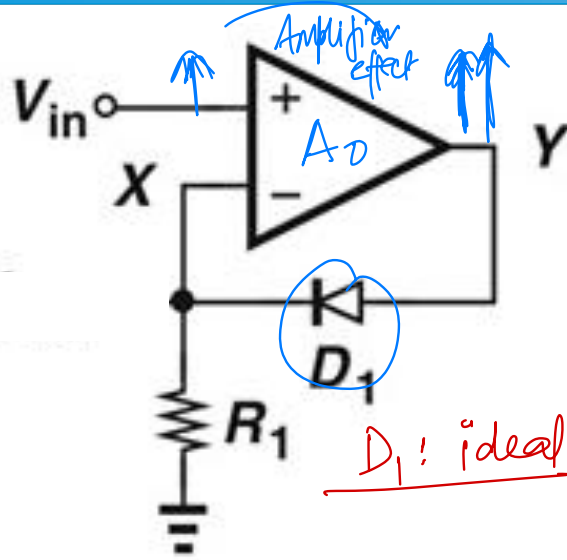
In open-loop :-



quite un-useful
for any amplifier application

if $V^+ \& V^- = \pm 15V$, $\& A_0 = 10^5 \Rightarrow \left. \begin{aligned} \frac{V^+}{A_0} &= 150 \mu V \\ \& \frac{V^-}{A_0} &= -150 \mu V \end{aligned} \right\}$

Non-inverting rectifier



$$V_{in} > 0 \rightarrow V_y \gg 0$$

$$D_1 \rightarrow \text{s.c.}$$

$$V_x = V_y = V_{in}$$

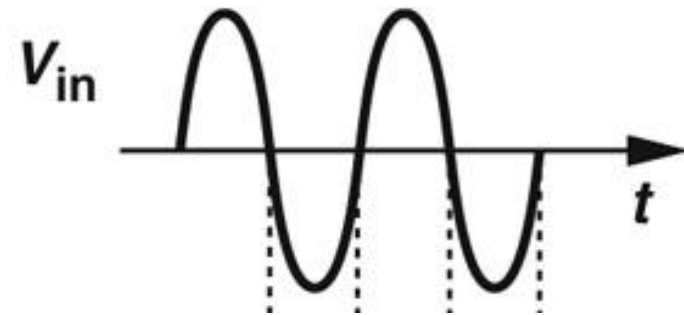
$$\text{if } V_{in} < 0 \rightarrow V_y \ll 0$$

$$\Rightarrow D_1 \rightarrow \text{o.c.}$$

$$V_x \sim 0 \text{ (no current in } R_1 \text{)}$$

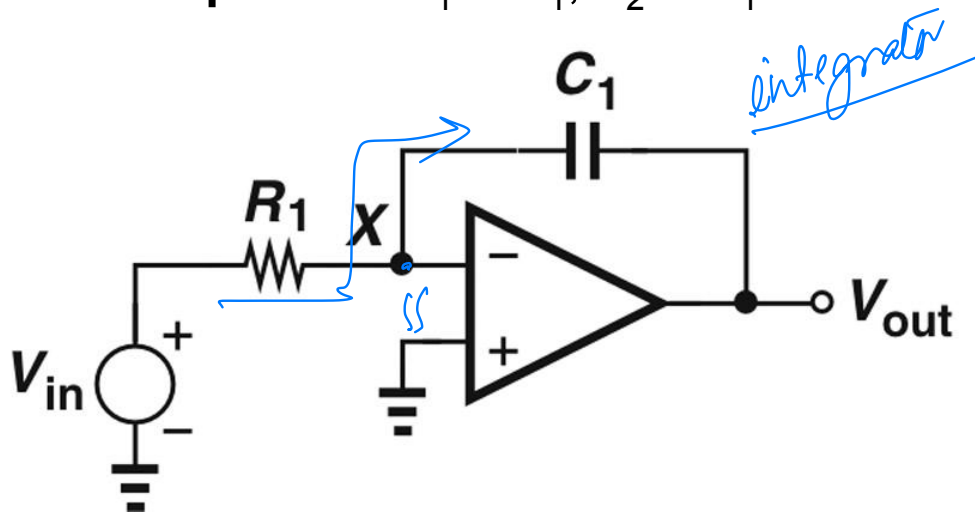
$$V_y = A_0 V_{in} \text{ (if } |V_{in}| < \frac{V_{cc}}{A_0} \text{)}$$

$$\text{if } |V_{in}| > \frac{V_{cc}}{A_0} \Rightarrow V_y = -V_{cc}$$



Two complementary functionalities with OpAmp

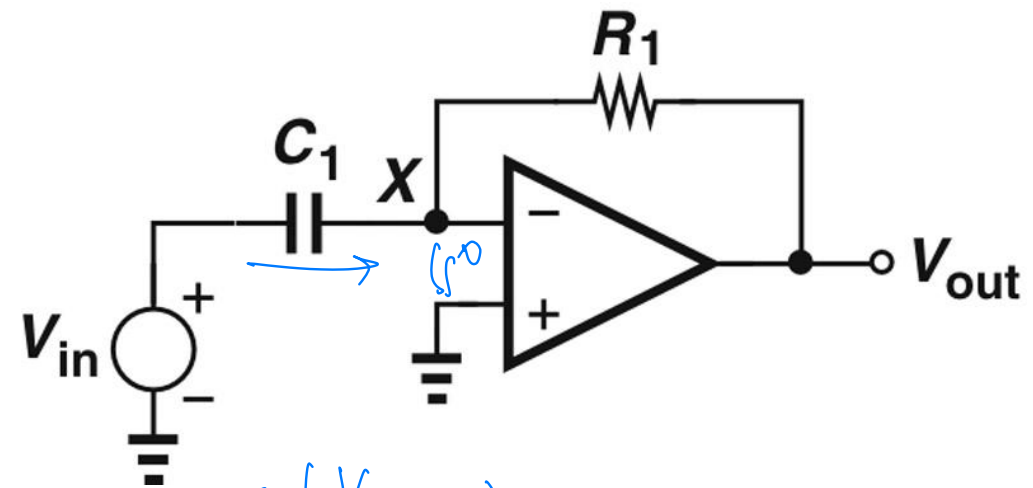
Option 1: $Z_1 = R_1, Z_2 = C_1$



$$\frac{V_{in} - 0}{R_1} = C_1 \left(\frac{0 - V_{out}}{dt} \right)$$

$$V_{out} = -\frac{1}{R_1 C_1} \int V_{in} dt$$

Option 2: $Z_1 = C_1, Z_2 = R_1$



$$C_1 \left(\frac{V_{in} - 0}{dt} \right) = \frac{0 - V_{out}}{R_1}$$

$$V_{out} = -R_1 C_1 \frac{dV_{in}}{dt}$$